

WR-VIII

The Cosmological Fibre: Universes Compatible with Our Universe Today

Distance responses, curvature completion
and global ambiguity
Mathematical draft v0.1

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Abstract

Cosmological responses constrain a declared model class before they identify a global universe. We construct an explicit FLRW kinematic carrier on a finite observed redshift interval. Every strictly increasing calibrated transverse-distance profile admits a one-parameter family of constant-curvature completions, with a different positive expansion history for each admissible curvature. An independently calibrated expansion-rate response selects the curvature within that carrier; inconsistent rates can make its fibre empty. Relative supernova distances and BAO ratios retain an absolute-scale ambiguity when their calibration scale is free. A three-parameter polynomial distance family gives a finite-dimensional benchmark with an exact fibre, an explicit rank calculation and sharp interval certificates. Even an identified flat background admits globally different spatial lifts: Euclidean space and sufficiently large cubic tori agree on the supplied classical observed patch. Smooth future extensions give a separate ambiguity when a matter evolution law is not supplied. Standard distance and curvature identities are attributed to established cosmology; no new gravity theory, observational fit or identification of the actual Universe is claimed. The bilingual package includes analytic figures, synthetic reproduction and an offline demonstration.

Keywords: FLRW; cosmological response; curvature completion; calibration; topology; model-relative identification.

Scope: kinematic background, complete EN/RU text, synthetic benchmarks. REVIEWABLE_DRAFT; no WR-VIII DOI or external peer review. Content CC BY 4.0; code MIT.

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1 The new obligation: a cosmological response carrier

WR-I supplies response equivalence and identifiable quantities [1]; WR-II separates finite fitting from global realization [2]; WR-III studies nonempty fibre geometry [3]; WR-IV separates observed history from its full lift [4]. WR-V adds experimental transitions [5]; WR-VI and WR-VII require independently justified physical constraints and calibrated resources [6, 7]. The WREH manifesto assigns this paper a cosmological obligation [8]. A quantum speed certificate alone does not supply a cosmological response map. We supply standard distance and expansion-rate responses of a declared homogeneous, isotropic background. “Today” identifies the observer’s information at a fixed reception time; it does not mean that the synthetic examples fit current surveys. The actual cosmological fibre needs an observational data vector, covariance or justified error set, nuisance parameters, survey selection and a specified matter/perturbation model. Those inputs are not invented here. Three objects remain separate: background parameters on the observed interval; a metric and fields on the observed patch; a global spatial/time extension. We compute the first fibre and construct explicit ambiguities of the latter lifts. Existing distance and curvature identities [9, 10] are used as established tools. The new programme deliverable is an audited completion contract and reproducible benchmark, without a priority claim for those tools.

2 Metric, branch and calibrated response record

Fix $Z > 0$, reception time $t_0 = 0$, $c > 0$, and $z \in [0, Z]$. A kinematic FLRW background has constant present curvature κ (length⁻²), positive $H \in C^1[0, Z]$ (time⁻¹), and scale-factor normalization $a(0) = 1$. With comoving radial length χ , its local metric and clock reconstruction are

$$ds^2 = -c^2 dt^2 + a(t)^2 [d\chi^2 + S_\kappa(\chi)^2 d\Omega^2],$$

$$t(z) = - \int_0^z \frac{du}{(1+u)H(u)}, \quad a(t(z)) = \frac{1}{1+z}. \quad (1)$$

$$\chi(z) = c \int_0^z \frac{du}{H(u)}, \quad D(z) = S_\kappa(\chi(z)),$$

$$S_\kappa(x) = \begin{cases} \sin(\sqrt{\kappa}x)/\sqrt{\kappa}, & \kappa > 0, \\ x, & \kappa = 0, \\ \sinh(\sqrt{-\kappa}x)/\sqrt{-\kappa}, & \kappa < 0. \end{cases} \quad (2)$$

For $\kappa > 0$ impose $\chi(Z) < \pi/(2\sqrt{\kappa})$, the strictly increasing branch of S_κ . This restriction concerns transverse comoving distance D , not the angular-distance turnover. It excludes conjugate/turning branches instead of silently continuing across them. The familiar curvature-density convention is $\Omega_k = -\kappa c^2 / H(0)^2$. Distance duality, standard propagation and independently justified luminosity/ruler calibration are supplied assumptions. Under them the background responses are [9]

$$D_L(z) = (1+z)D(z), \quad D_A(z) = \frac{D(z)}{1+z}, \quad R_H(z) = H(z), \quad D_H(z) = \frac{c}{H(z)}. \quad (3)$$

An exact profile here is an ideal calibrated function, including its derivative. Finitely many distance samples do not supply D' unless a justified interpolation/model family is declared. A kinematic metric is not automatically an admissible solution for a prescribed matter equation of state, energy condition or perturbation spectrum. Each such condition defines a smaller, separately checked carrier.

Definition 2.1 (Observed-interval completion). *For a protocol set $\mathcal{A}$ and response record $\mathcal{D} = (J_P)_{P \in \mathcal{A}}$, define the kinematic fibre in the above carrier $\mathcal{W}_{\text{kin}}$ by*

$$\mathfrak{C}_{\text{kin}}(\mathcal{D}) = \{(\kappa, H) \in \mathcal{W}_{\text{kin}} : R_P(\kappa, H) \in J_P \text{ for every } P \in \mathcal{A}\}. \quad (4)$$

A spatial topology, unobserved epoch or matter decomposition is an additional lift, with its own admissibility and response rules.

3 The exact distance fibre and curvature selection

Theorem 3.1 (All first-branch distance completions). *Let $D \in C^2[0, Z]$, $D(0) = 0$, and $D'(z) > 0$ throughout. In the carrier of Section 2, the entire exact distance fibre is*

$$\{(\kappa, H_\kappa) : \kappa \in K_D\}, \quad K_D = \left(-\infty, \frac{1}{D(Z)^2}\right), \quad H_\kappa(z) = \frac{c\sqrt{1 - \kappa D(z)^2}}{D'(z)}. \quad (5)$$

In particular $H_\kappa(0) = c/D'(0)$ for every member.

Proof. On the declared branch $S'_\kappa(x) > 0$ and $S'_\kappa(x)^2 = 1 - \kappa S_\kappa(x)^2$. Differentiate (2) to obtain

$$D'(z) = \frac{c}{H(z)} \sqrt{1 - \kappa D(z)^2}. \quad (6)$$

Positivity at Z forces $\kappa < D(Z)^{-2}$, and solving gives the unique H_κ for fixed κ . Conversely, for any $\kappa \in K_D$ let $F_\kappa(y) = \int_0^y (1 - \kappa s^2)^{-1/2} ds$. Then $\chi(z) = F_\kappa(D(z))$ has derivative $c/H_\kappa(z)$ and initial value zero. F_κ is the inverse of the chosen branch of S_κ ; for positive κ its value at $D(Z)$ is strictly below $\pi/(2\sqrt{\kappa})$. Thus (2) holds on the whole interval. H_κ is positive and C^1 , and (1) defines the scale factor on its reconstructed time interval. $\square$

This is an explicit completion formulation of standard FLRW geometry [9, 10]. The open endpoint is essential: at equality the reconstructed $H(Z)$ vanishes while the carrier requires $H > 0$. Distinct κ give distinct present spatial curvature and distinct $H(z)$ for $z > 0$, although every distance response in (3) agrees.

Corollary 3.2 (Independent rates select a background or exclude the carrier). *For the exact D of Theorem 3.1, one independent exact rate $h_* > 0$ at $0 < z_* \leq Z$ selects at most one member. Its candidate curvature is*

$$\hat{\kappa}(z_*) = \frac{1 - [h_* D'(z_*)/c]^2}{D(z_*)^2}. \quad (7)$$

The member exists exactly when $\hat{\kappa} \in K_D$. Several such rates have a common completion exactly when their candidates agree and belong to K_D .

Proof. Equation (6) fixes (7). Conversely, substituting an admissible candidate into (5) reproduces each supplied positive rate; Theorem 3.1 supplies the common distance profile. $\square$

The curvature estimator is the Clarkson–Bassett–Lu identity [10], expressed in dimensional κ . It is not a new curvature test. Their geometric consistency function becomes

$$\mathcal{C}(z) = 1 + \frac{H^2}{c^2}(DD'' - D'^2) + \frac{HH'}{c^2}DD', \quad \hat{\kappa}'(z) = -\frac{2D'(z)}{D(z)^3}\mathcal{C}(z), \quad z > 0. \quad (8)$$

Thus $\mathcal{C} = 0$ enforces a constant candidate on this branch. Passing the test identifies a background only within the declared carrier; it does not prove global FLRW symmetry among all spacetime models. A rate inferred from the same distance data after assuming curvature is not an independent selecting response. Derivatives and their errors need independent justification.

4 Calibration is part of the fibre

Real distance channels carry nuisance scales. For an ideal common supernova absolute magnitude M and a free BAO ruler $r_d > 0$, write the background observables as

$$m(z) = M + 5 \log_{10} \frac{D_L(z)}{10 \text{ pc}}, \quad R_\perp(z) = \frac{D(z)}{r_d}, \quad R_\parallel(z) = \frac{c}{H(z)r_d}. \quad (9)$$

These expressions are background interfaces, not a survey likelihood. BAO analyses report such distance/ruler ratios with model and fiducial-processing assumptions [12]; a calibrated ratio does not supply a calibrated ruler length.

Proposition 4.1 (A joint scale ambiguity). *If M and r_d are free, then for every $\lambda > 0$ the transformation*

$$(D, H, \kappa, r_d, M) \mapsto (\lambda D, H/\lambda, \kappa/\lambda^2, \lambda r_d, M - 5 \log_{10} \lambda) \quad (10)$$

preserves the metric-distance relation and all responses in (9). It changes the absolute length/time scale; an independently calibrated absolute rate or ruler can constrain it.

Proof. The new radial integral is $\lambda\chi$, and $S_{\kappa/\lambda^2}(\lambda\chi) = \lambda S_{\kappa}(\chi)$. The first-branch inequality is unchanged. Both BAO ratios cancel λ , while the magnitude shift cancels the change of D_L . An absolute rate transforms as H/λ and therefore is not preserved. $\square$

Absolute-scale ambiguity and curvature/expansion ambiguity are different. With freely calibrated scales, “more channels” need not provide the missing absolute input. This is an application of WR-I’s response contract; the scaling symmetry is not claimed as a new cosmological discovery.

5 A finite-dimensional cosmological benchmark

Take the three-parameter kinematic family (u, v, κ) with $u > 0$, $\min\{u, u + 2vZ\} > 0$, and $\kappa < (uZ + vZ^2)^{-2}$. Both u and v are lengths. The profiles are

$$D(z) = uz + vz^2, \quad H(z) = \frac{c\sqrt{1 - \kappa(uz + vz^2)^2}}{u + 2vz}. \quad (11)$$

The polynomial is an exact declared benchmark over $[0, Z]$, not a cosmographic truncation whose omitted terms are ignored. Its use for measured data would need its own model-error assessment.

Theorem 5.1 (Two distance samples fix shape, not curvature). *For $0 < z_1 < z_2 \leq Z$, exact distances $y_i = D(z_i) > 0$ fix*

$$v = \frac{y_2/z_2 - y_1/z_1}{z_2 - z_1}, \quad u = \frac{z_2 y_1/z_1 - z_1 y_2/z_2}{z_2 - z_1}. \quad (12)$$

If the resulting u, v satisfy the declared positivity conditions, the distance fibre is exactly the open interval $\kappa < (uZ + vZ^2)^{-2}$; otherwise it is empty. Adding one exact independent $H(z_)$, $0 < z_* \leq Z$, gives a locally full-rank three-parameter response map and at most one admissible member.*

Proof. Divide y_i by z_i and solve the two affine equations. Their Jacobian determinant is $z_1 z_2 (z_2 - z_1) \neq 0$. They do not depend on κ , whose entire admissible interval is supplied by Theorem 3.1. For the third response,

$$\frac{\partial H(z_*)}{\partial \kappa} = -\frac{cD(z_*)^2}{2D'(z_*)\sqrt{1 - \kappa D(z_*)^2}} < 0, \quad \det DR = z_1 z_2 (z_2 - z_1) \frac{\partial H(z_*)}{\partial \kappa} \neq 0. \quad (13)$$

The triangular response Jacobian has rank three, and Corollary 3.2 gives the stronger global at-most-one statement in this family. $\square$

Rank is conditional on the exact finite model and absolute calibration. This is not identification of all expansion histories from two distance samples. At least the residual global lifts below remain.

6 Sharp rate-interval certificates and synthetic units

Proposition 6.1 (Exact projected curvature interval). *Keep an exact D satisfying Theorem 3.1. At finitely many $z_i > 0$, supply independent admissible rate intervals $0 < h_i^- \leq h_i^+$. The projected compatible curvature set is exactly*

$$K_{\mathcal{D}} = K_D \cap \bigcap_i [\kappa_i^-, \kappa_i^+], \quad \kappa_i^- = \frac{1 - (h_i^+ D'_i/c)^2}{D_i^2}, \quad \kappa_i^+ = \frac{1 - (h_i^- D'_i/c)^2}{D_i^2}. \quad (14)$$

The fibre is empty, a singleton, or an interval precisely as this intersection is. An additional independently justified hard prior intersects the same set.

Proof. For $z_i > 0$, $H_\kappa(z_i) > 0$ is strictly decreasing in κ . Solving $h_i^- \leq H_\kappa(z_i) \leq h_i^+$ gives the displayed closed interval. A common $\kappa \in K_D$ generates the exact D and all admitted rates by Theorem 3.1, so the intersection is sufficient as well as necessary. $\square$

These are deterministic admissibility intervals. Their confidence or joint coverage is not supplied by the algebra. Correlated measurements need a joint error set. If distances or derivatives are uncertain too, varying them independently gives at most an outer necessary screen until one jointly realizable distance profile is constructed. Failure excludes the supplied carrier/calibration combination. For a synthetic unit-bearing benchmark set $c = 299\,792.458$ km/s, $H_0 = 70$ km/s/Mpc, $Z = 1.5$, and

$$u = \frac{c}{H_0} = 4282.7494 \text{ Mpc}, \quad v = -0.15u, \quad f(z) = z - 0.15z^2, \quad (15)$$

$$q = \kappa u^2, \quad \frac{H_q(z)}{H_0} = \frac{\sqrt{1 - qf(z)^2}}{1 - 0.30z}.$$

Every admissible q has the same $D = uf$ and $H(0) = H_0$. The branch bound is $q < f(1.5)^{-2} \approx 0.739970$. At $z = 1$ the synthetic flat member has $H = 100$ km/s/Mpc. A rate interval $[98, 102]$ gives

$$q \in \left[\frac{1 - 1.02^2}{0.85^2}, \frac{1 - 0.98^2}{0.85^2} \right] \approx [-0.05591696, 0.05480969]. \quad (16)$$

An exact rate 100 selects $q = 0$ within this background family. For example, an exact second rate $H(0.5) = 100$ instead gives $q = (1 - (0.85 \cdot 100/70)^2)/0.4625^2 < 0$, so it cannot coexist with that exact flat rate at $z = 1$. No actual catalogue is represented by these witnesses.

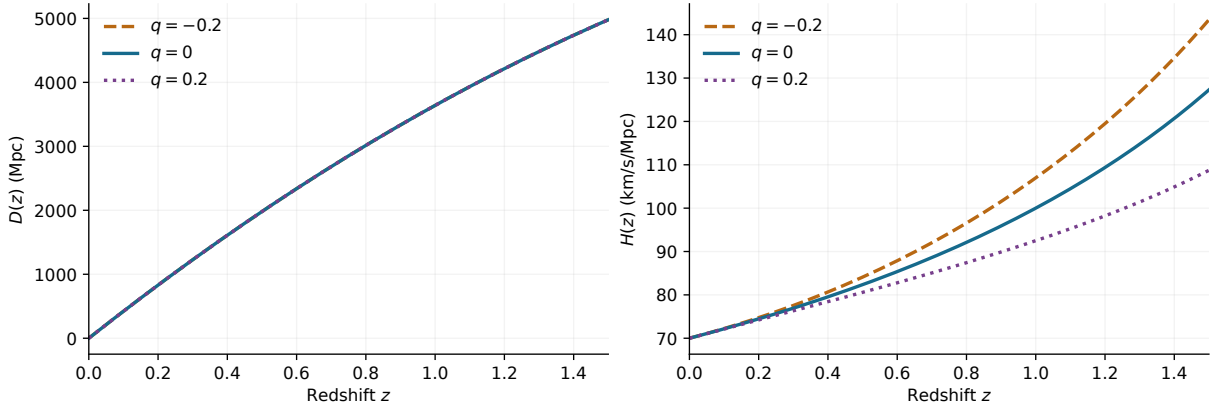


Figure 1: Analytic synthetic curves: identical calibrated distances and distinct expansion histories for $q = -0.2, 0, 0.2$. Units are Mpc and km/s/Mpc; no measurements are plotted.

7 Global spatial lifts of a flat background

The topology question is separately declared, in the established cosmic-topology setting [11]. Fix a positive scale factor on a time interval I , normalized by $a(0) = 1$, and compare the flat spatial sections $\mathbb{R}^3$ and the cubic torus $T_L^3 = \mathbb{R}^3/(L\mathbb{Z})^3$ with the same local metric $-c^2 dt^2 + a(t)^2 d\mathbf{x}^2$. The torus is locally FLRW; global isotropy about every point is not imposed as an additional topology condition.

Theorem 7.1 (A finite observed patch does not fix flat topology). *Suppose the supplied classical protocols depend only on the metric and copied classical source fields in $I \times \overline{B_R(0)}$, and all used light paths lie in that patch. For each $L > 2R$, Euclidean space and T_L^3 admit identical protocol responses on this patch, while their spatial topologies and total volumes differ. In the flat distance benchmark one may take $R = \chi(Z) = D(Z)$.*

Proof. If $\mathbf{x}, \mathbf{y} \in \overline{B_R}$ have the same torus image, then $\mathbf{x} - \mathbf{y} = L\mathbf{n}$ for an integer vector $\mathbf{n}$. For $\mathbf{n} \neq 0$, its norm is at least $L > 2R$, contradicting $\|\mathbf{x} - \mathbf{y}\| \leq 2R$. Hence the quotient is injective on the ball and is a local isometry there; its time-product restriction preserves the metric and the copied fields. Every stipulated protocol functional therefore agrees. Spatial $\mathbb{R}^3$ is noncompact and simply connected, whereas T_L^3 is compact with fundamental group $\mathbb{Z}^3$ and present volume L^3 . Homogeneous classical fields provide explicit global copies on both spaces. $\square$

$$L > 2\chi(Z) \implies \mathbb{R}^3 \text{ and } T_L^3 \text{ share the declared observed patch,} \quad V(T_L^3, t_0) = L^3. \quad (17)$$

This proves a witness for the stipulated local classical responses. A stochastic perturbation prior, global eigenmode spectrum or quantum state can carry additional topology-dependent information and requires another response model. Failing $L > 2R$ only removes this sufficient witness; it does not prove detectability. The restriction is finite and protocol-dependent, not an in-principle equivalence for every future observer. WR-IX owns that stronger accessibility question.

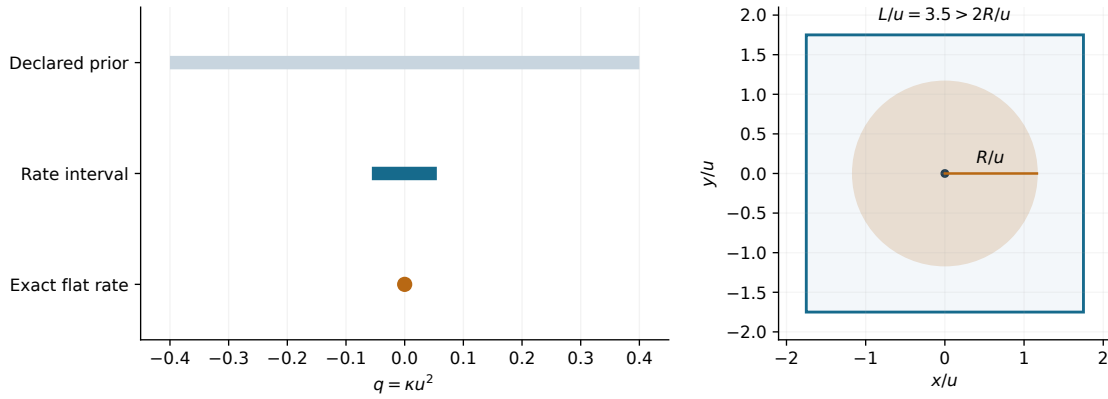


Figure 2: Left: synthetic curvature-prior interval, refinement by a rate interval, and the exact flat rate. Right: a two-dimensional cross-section illustrating the injective ball inside a cubic torus domain; the full construction is three-dimensional. No cosmological data fit is shown.

8 A separate temporal lift

Proposition 8.1 (Past agreement permits different kinematic futures). *Let $a_b > 0$ be C^∞ on $I = (-T_-, T_+)$, $a_b(0) = 1$, and $\mathcal{H}_b = \dot{a}_b/a_b > 0$. Choose a nonzero smooth dimensionless bump b with compact support in $(0, T_+)$. Then a nontrivial interval of λ gives*

$$a_\lambda(t) = a_b(t)e^{\lambda b(t)}, \quad \mathcal{H}_\lambda(t) = \mathcal{H}_b(t) + \lambda \dot{b}(t) > 0, \quad a_\lambda(t) = a_b(t) \quad (t \leq 0). \quad (18)$$

These histories agree in all supplied past-patch responses and differ in the future.

Proof. The bump vanishes together with all derivatives near $t = 0$ and outside its compact support. Positivity and smooth matching follow immediately. Let $h_{\min} > 0$ be the minimum of $\mathcal{H}_b$ on the support and $B = \|\dot{b}\|_\infty > 0$. For $|\lambda| < h_{\min}/B$, the new expansion rate stays positive there and equals the original elsewhere. Distinct λ give distinct scale histories where $b \neq 0$, with the same normalization and anchored clock. Past metric and copied source responses are unchanged. $\square$

This is a conditional kinematic lift. A supplied matter evolution law and complete initial data may enforce a unique future or exclude the bump histories. No family of solutions to one fixed matter law is asserted. WR-IV's distinction between a projection and a full history is applied, rather than re-proved as a cosmological law.

9 What is supplied and what a data analysis still needs

Channel/object	Supplied here	Separate obligation
Supernova distances	D_L , ideal magnitude and free zero point	Catalogue, populations, dust, selection, covariance, absolute calibration
BAO	$D/r_d, c/(Hr_d)$	Ruler physics, fiducial processing, covariance and joint likelihood
Independent rates	Exact or deterministic intervals	Clock model, calibration, derivative control, correlated errors
CMB, lensing, growth	No full response engine	Perturbations, primordial fields, transfer functions and observations
Global lift	Explicit flat topology and kinematic future witnesses	Matter admissibility, global fields, full-time extension and broader protocols

An empty $\mathfrak{C}_{\text{kin}}$ rejects a declared model/data/calibration combination; it is not the WR-II statement that no full physical world can realize the observations. A singleton (κ, H) is an identified background on a supplied interval; Sections 7–8 exhibit residual lifts. A curvature-admissibility endpoint is a branch/model boundary; no spacetime edge or WR-III structural wall is inferred without its own total-family analysis. The present paper opens the cosmological interface and leaves the actual combined CMB/BAO/SN/lensing fibre as a documented next work package. Its finite benchmark introduces a physical response map, rather than renaming the nonnegative transition inverse problem of WR-V. Its classical quotient witness is separate from WR-VII’s quantum generator circle. WREH remains both a research programme and community, distinct from Boundary Compensation; WR-IX remains planned.

10 Reproduction, failure criteria and review gate

Reproduction independently integrates c/H_κ and applies the curved distance function, checks the finite response Jacobian, interval inversion and incompatible-rate witnesses, verifies the joint calibration transformation, and checks lattice aliases and future matching. The finite checks accompany the seven proof blocks; they are not their premises. Figures and browser outputs are analytic/synthetic. The offline demonstration exposes curvature selection, interval refinement, the open branch cap and the sufficient torus-patch condition. Failure criteria are an undeclared distance/ruler calibration, a derivative inferred without error control, circular use of H , a closed-geometry branch crossing, a positivity violation, a hidden matter prior, or a universal topology/horizon inference from a restricted patch. A covariance likelihood cannot silently become a hard exact interval. Existing source identities do not establish novelty; their relevant primary sections and access status are recorded separately. WR-VII publication DOI 10.5281/zenodo.23264253 was reported by the author on 9 October 2026. Direct record and file verification was unavailable during this preparation and is a separate pending check; the frozen v0.2 source is cited explicitly. No DOI is assigned to WR-VIII. Status: REVIEWABLE_DRAFT; independent scientific review, actual survey inference and matter-model admissibility remain pending.

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